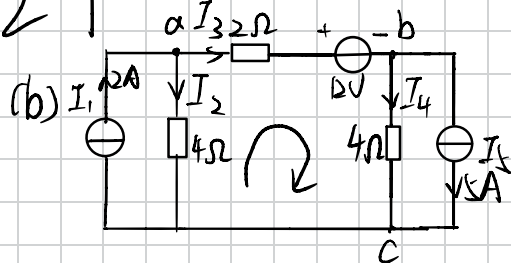


习题 2

2-1



3个节点 5条支路

由KCL定律

由节点 a, $I_1 - I_2 - I_3 = 0 \dots ①$

由KVL定律

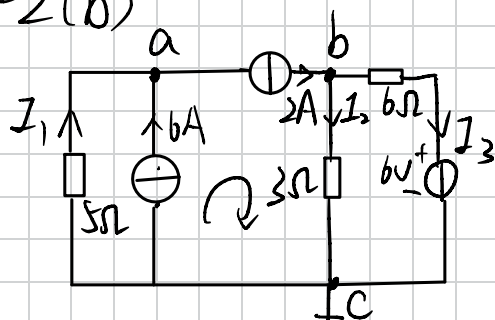
由节点 b, $I_3 - I_4 - I_5 = 0 \dots ②$

$-4I_2 + 2I_3 + 12 + 4I_4 = 0 \dots ③$

联立 ① ② ③ 得

$$\begin{cases} I_2 = 0.4A \\ I_3 = 1.6A \\ I_4 = -3.4A \end{cases}$$

2-2(b)



3个节点 5条支路

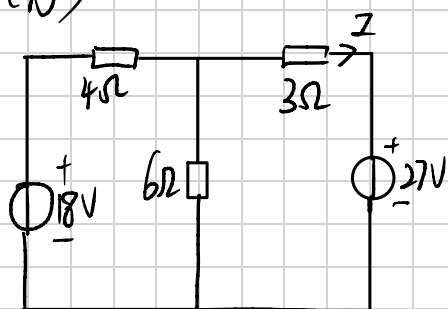
设节点 a, b 电位 V_a, V_b , 由于 c 节点接地所以 $V_c = 0$

$$I_1 = \frac{V_a}{5} \quad I_2 = \frac{V_b}{3} \quad I_3 = \frac{V_b - 6}{6}$$

由节点 a, b 列 KCL 方程

$$\begin{cases} \frac{V_a}{5} + 6 - 2 = 0 \\ 2 - \frac{V_b}{3} - \frac{V_b - 6}{6} = 0 \end{cases} \Rightarrow \begin{cases} V_a = 20V \\ V_b = 6V \end{cases}$$

2-3(b)



当 18V 电压源单独作用时

将 27V 电压源作短路处理

$$I_{\text{总}} = \frac{18}{4 + \frac{3 \times 6}{3 + 6}} = 3A$$

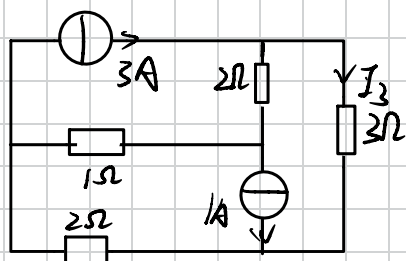
$$I' = 3 \times \frac{6}{3 + 6} = 2A$$

当27V电压源单独作用时,将18V电压源为短路处理

$$I_{\text{总}} = \frac{27}{3 + \frac{6 \times 4}{6+4}} = 6A = -I''$$

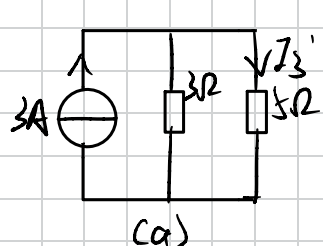
$$\therefore I = I' + I'' = 2 - 10 = -8A$$

2-4



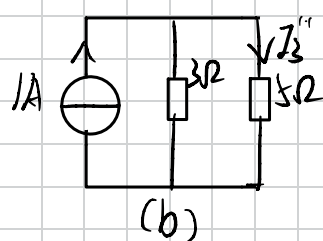
当3A电流源单独作用时将1A电流源作断路处理

则电路图可画作下图(a)



$$I_3' = 3 \times \frac{3}{5+3} = \frac{9}{8} A$$

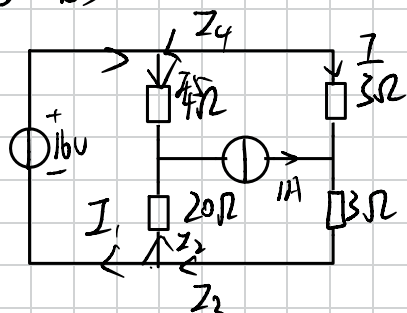
当1A电流源单独作用时,将3A电流源作断路处理
则电路可画作下图(b)



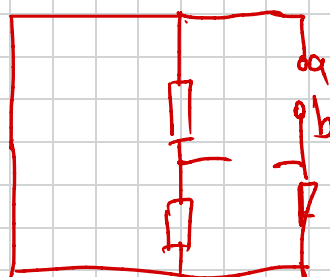
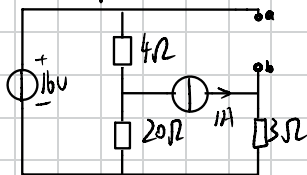
$$I_3'' = 1 \times \frac{3}{5+3} = \frac{3}{8} A$$

$$\therefore I_3 = I_3' + I_3'' = \frac{12}{8} = 1.5A$$

2-6(b)



断开电流I所在支路如图



$$\therefore U_{ab} = 16 - 3 \times 1 = 13V$$

由于20Ω和4Ω电阻被短路

等效电阻为3Ω

$$\therefore I = \frac{13}{3+3} \approx 2.167 A$$